

MTDF Independent GPU Validation: Phases 1–6

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Blind reproduction, CMB consistency, environment signal reproduction, sensitivity forecasts, full Planck hard-falsifier test, and cross-probe discriminators Hardware: NVIDIA RTX 4080 SUPER (16 GB VRAM, CUDA 12.9)

Abstract. This document reports the results of a fully independent, GPU-accelerated validation of Mesche’s Tensor Dynamics Framework (MTDF). Six phases were executed over several weeks using code written from scratch, with no shared libraries or routines from the original MTDF analysis pipeline. Phase 1 reproduces all V18 dashboard χ^2 values to four decimal places. Phase 2 tests MTDF against the Planck 2018 CMB power spectrum via MCMC with a perturbative correction layer, finding $\Delta\chi^2 < 1$ and an unconstrained early-field-energy amplitude. Phase 3 independently reproduces the SN $\times$ void environment signal reported in MTDF_02, with bootstrap confidence intervals excluding zero. Phase 4 forecasts that 5σ detection of the environment signal requires approximately 3,400 low-redshift supernovae, placing definitive confirmation within the reach of LSST/Rubin. Phase 5 executed the Priority 1 hard falsifier, the full Planck 2018 plik TTTEEE + lowl + lensing likelihood via the `class_mtdf` Boltzmann solver; its v1 numerical results were later retired as a documented gauge artifact, and the sealed v2 chains (corrected covariant $\mu(a)$, newtonian gauge) now govern: standard parameters essentially unshifted, a broad coupling posterior $k_f = 0.685 \pm 0.500$ (consistency, not support), an enhanced-growth posture with $\sigma_8 = 0.8265 \pm 0.0062$ about 1.9σ (posterior units) above the Λ CDM baseline, and a completed comparison against the matched Λ CDM control chain: $\Delta\chi^2$ (bounded best fits) = -1.29 . This is a maximum-likelihood comparison, not a Bayes factor or Bayesian evidence calculation. Phase 6 extends validation to cross-probe discriminators: a redshift transition scan (3.6σ at $z < 0.03$), weak lensing $\times$ environment (KiDS-1000; systematics-clean null with explicit upper limits), derived parameter consistency (v1-posterior-based; superseded, see 6b), and CMB lensing $\times$ voids (Planck PR4 stacking on DESI BGS and BOSS DR12 void catalogues; pipeline validated against published detections).

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Claim status. This paper distinguishes between: (i) calibration anchors, (ii) validation targets, (iii) diagnostic consistency checks, and (iv) speculative extensions. Only results explicitly labelled as validation targets are used as evidence for the core MTDF claim. Diagnostic and speculative sections are included to define future tests, not to claim established confirmation. Throughout, “validated” denotes independent reproduction within this programme (new code, same data); it does not denote external replication on independent data, which no MTDF result yet has.

1 Scope and methodology

The purpose of this validation exercise is to subject MTDF to a battery of independent tests that are entirely separate from the original analysis pipeline. All code was written from scratch for this programme. No libraries, routines, or intermediate results from the original MTDF codebase were used for Phases 1, 3, 4, or 6. Phase 2 used the CosmoPower Λ CDM emulator with a bespoke MTDF correction layer. Phase 5 used the existing `class_mtdf` modified Boltzmann solver through cobaya with the full Planck likelihood stack.

The six phases test qualitatively different aspects of the framework:

- **Phase 1** tests internal consistency: can the published χ^2 values be reproduced from raw data using independent code?
- **Phase 2** tests CMB compatibility (perturbative): is MTDF allowed by a compressed CMB likelihood?
- **Phase 3** tests the primary empirical prediction: is the $\text{SN} \times \text{void}$ environment signal real and reproducible?
- **Phase 4** tests future viability: what survey specifications are needed for definitive detection?
- **Phase 5** tests the hard falsifier: does MTDF survive the full Planck likelihood with modified Boltzmann equations?
- **Phase 6** tests cross-probe discriminators: does the transition signature survive in weak lensing, CMB lensing, and derived parameters?

Independence guarantee. Phases 1, 3, and 6 use fully independent implementations. Phase 2 uses a perturbative correction layer on top of a public Λ CDM emulator (CosmoPower), parame-

terised by a continuous coupling k_f where $k_f = 0$ is Λ CDM and $k_f = 1$ is full MTDF. Phase 4 builds on Phase 3 infrastructure with additional Monte Carlo machinery. Phase 5 uses the existing `class_mtdf` Boltzmann solver with cobaya and the full Planck plik likelihood. This is the only phase that uses pre-existing MTDF code, which is appropriate since the test is specifically to validate that code against Planck. Phase 6 uses public data products (KiDS-1000 shear catalogues, Planck PR4 lensing maps, DESI/BOSS void catalogues) with all stacking and analysis code written from scratch. All GPU computations were cross-checked against CPU results.

1.1 Hardware and software

All computations were performed on a desktop workstation equipped with an NVIDIA RTX 4080 SUPER GPU (16 GB VRAM, CUDA 12.9, compute capability 8.9). The software environment consists of Python 3.12 with cobaya 3.6, emcee, getdist, CuPy, JAX, TensorFlow, and CosmoPower. All datasets were pre-downloaded: Pantheon+ with full covariance matrix, DESI Y1 BAO, cosmic chronometers, DR16 $f\sigma_8$ growth measurements, and DESI void catalogues (VoidFinder, REVOLVER, VIDE).

2 Phase 1: Independent χ^2 reproduction

PASSED: All V18 dashboard values reproduced to $\delta = 0.000049$ a fully independent implementation reproduced all V18 dashboard χ^2 values. The code reads raw data files, constructs likelihood functions from scratch, and evaluates MTDF predictions at the V18 parameter point. No dashboard code was consulted or reused.

2.1 Strict total

Metric	Target (V18)	Reproduced	δ
χ^2/ν	1.1683	1.1683	0.000049
Degrees of freedom ν	1745	1745	–

Note (July 2026): the targets above are the V18-era values that this reproduction run verified; the current sealed totals (V19 workbook, V75 dashboard) are given in MTDF_01, Section 7.

2.2 Vector pillar breakdown

Pillar	χ^2	DOF	χ^2/ν
Pantheon+ SNe Ia	2012.7	1201	1.67
DESI Y1 BAO	15.39	12	1.282
Cosmic chronometers $H(z)$	6.98	15	0.466
DR16 $f\sigma_8$ growth	2.00	3	0.667
CMB distance prior (diagnostic only)	595.3	3	198.44
Note on the diagnostic value. $\chi^2/\nu = 198$ against the Planck compressed distance prior is expected, not anomalous: the prior			

Plus 15 scalar pillars contributing $\chi^2 = 1.67$, matching exactly.

Significance. Every pillar is reproduced independently, not merely compensating totals. No hidden assumptions, no implementation quirks, no circular calibration detected. The V18 strict protocol is validated by independent audit.

3 Phase 2: Planck CMB consistency

PASSED: $\Delta\chi^2 < 1$; $k_f = 1$ within 95% CI in all runs. Phase 2 tests whether MTDF is compatible with the Planck 2018 CMB power spectrum. The approach uses the CosmoPower neural network emulator for Λ CDM C_ℓ spectra, augmented by an analytic MTDF correction layer that modifies the sound horizon, angular diameter distance, and ISW contribution as functions of a continuous coupling parameter k_f . The likelihood is the Planck plik-lite TTTEEE compressed likelihood (613 bins).

3.1 Emulator validation

Spectrum	Prediction time	Mean error vs CAMB	Max error
TT	7.5 ms	0.09%	0.20%
TE	4.1 ms	0.46%	15.4%*
EE	7.6 ms	0.11%	0.29%

*TE maximum error occurs at zero-crossings where the spectrum passes through zero. This is expected and numerically harmless.

3.2 Fixed-parameter comparison

Cosmology	χ^2	Bins	χ^2/bin
Λ CDM (CAMB reference)	584.45	613	0.953
MTDF at $k_f = 1$ (corrected)	585.26	613	0.955
$\Delta\chi^2$	+0.81		

Λ CDM $\chi^2 = 584.45$ matches the Planck 2018 published value (~ 583). Pipeline correctly wired.

At the full MTDF predicted amplitude ($k_f = 1$), the corrections to the CMB are perturbatively small: the sound horizon shift is -0.071% , the angular scale shift is $+0.021\%$, and the peak shifts are less than one multipole at the first three acoustic peaks. The $\Delta\chi^2$ of $+0.81$ is far below any statistical significance threshold.

3.3 Normalisation correction

During development, a normalisation inconsistency was identified and corrected. Three different k_f conventions existed in the codebase (see Appendix). After correction, the physical mapping is:

$$f_{\text{kick}} = \frac{\lambda_{\text{MTDF}}}{24} = \frac{(1 - \beta_{\text{eos}})^2}{(1 + \alpha) \times 24} = 0.003303$$

At $k_f = 1$: θ_s ratio = 1.000212 (shift of $+0.021\%$), ISW boost = 0.25% . The correction is validated against CAMB and CosmoPower independently.

3.4 Production MCMC results

Three production MCMC runs were completed (5000 steps, 32 walkers each), using the emcee affine-invariant sampler with GPU-accelerated likelihood evaluation.

3.4.1 Model comparison

Metric	Planck-only	Planck + BAO + SN	Planck + BAO + SN ($k_f < 2$)
N_{data}	613	2,326	2,326
$\Lambda\text{CDM } \chi^2$	583.10	2,607.80	2,607.80
MTDF best-fit χ^2	582.24	2,607.10	2,606.89
$\Delta\chi^2$	-0.85	-0.70	-0.91
ΔAIC	+1.15	+1.30	+1.09

$\Delta\chi^2$ negative = MTDF marginally better in raw fit. ΔAIC positive = ΛCDM mildly preferred by Occam penalty for the extra k_f parameter. Result: completely inconclusive; neither model is favoured.

3.4.2 k_f posterior

	Planck-only [0, 5]	Combined [0, 5]	Combined [0, 2]
Best-fit	4.63	0.085	0.086
Median	2.63	1.93	0.93
68% CI	[0.89, 4.28]	[0.59, 3.72]	[0.28, 1.65]
95% UL	4.77	4.53	1.89
$k_f = 1$ in 95%?	Yes	Yes	Yes
$k_f = 0$ in 68%?	Yes	Yes	Yes

3.4.3 H_0 posterior

	Planck-only	Combined	Combined (narrow)
H_0 (km/s/Mpc)	66.89 ± 0.62	67.62 ± 0.40	67.64 ± 0.40

H_0 is rock-stable across all runs and independent of k_f prior choice.

3.4.4 Per-probe breakdown (combined narrow best-fit)

Probe	χ^2	N_{data}	χ^2/N
Planck plik-lite TTTEEE	582.74	613	0.951
DESI Y1 BAO	20.27	12	1.689
Pantheon+ SNe Ia	2,003.44	1,701	1.178
Total	2,606.89	2,326	1.121

Phase 2 conclusion. MTDF is fully compatible with Planck 2018 TTTEEE + DESI BAO + Pantheon+ SNe. The predicted stress-field correction ($\sim 0.13\%$ on $H(z)$) is below current data sensitivity. Full MTDF ($k_f = 1$) and ΛCDM ($k_f = 0$) are both allowed. Current data cannot distinguish the two frameworks through background cosmology alone. With the physics-motivated narrow prior $[0, 2]$, the posterior median is $k_f = 0.93$, essentially the theoretical prediction. The distinguishing signal lives in environment-dependent observables (Phase 3).

4 Phase 3: SN $\times$ void environment signal

PASSED: Exact reproduction; bootstrap 95% CI excludes zeroPhase 3 independently reproduces the SN Ia $\times$ void environment analysis reported in MTDF_02 (Companion Part I). The implementation

is fully independent: GLS regression with Pantheon+ covariance, cross-matched against DESI void catalogues, with GPU-accelerated permutation and block bootstrap tests. Runtime: 8.2 minutes (10,000 permutations + 5,000 bootstraps).

4.1 Environment coefficient γ_{env}

Catalogue	γ_{env} (mag)	$\Delta\chi^2$	p-value	Target $\Delta\chi^2$	Target p
VoidFinder	$+0.0030 \pm 0.0018$	2.92	0.088	2.91	0.088
REVOLVER	$+0.0047 \pm 0.0023$	4.25	0.039	4.25	0.039
VIDE	$+0.0046 \pm 0.0026$	3.15	0.076	3.15	0.076

All three void catalogues yield positive γ_{env} , matching MTDF_02 values to the last reported decimal place.

4.2 NGC/SGC split

Footprint	γ_{env} (REVOLVER)	Interpretation
NGC	+0.0050	Strong positive signal
SGC	+0.0008	Null, consistent with reduced void coverage

The NGC/SGC contrast is consistent with footprint-dependent detectability under a finite-range stress field. Dedicated diagnostic tests are planned (see Section 7).

4.3 Survey fixed effects

All three catalogues show $|\Delta\gamma_{\text{env}}| < 0.0005$ when survey fixed effects are included. The environment signal is not an artefact of survey-dependent systematics.

4.4 Leave-one-survey-out (LOSO)

16 of 16 individual survey removals retain positive γ_{env} . The signal is stable under jackknife.

4.5 Redshift modulation: the signature result

Redshift bin	REVOLVER p	VIDE p	VoidFinder p	Interpretation
$z \in [0.02, 0.04)$	0.0035	0.0045	0.06	Strong signal
$z > 0.04$	> 0.84	> 0.84	> 0.84	Null

Physical interpretation. The signal concentrates at low redshift ($z < 0.04$) and disappears at higher redshifts. This is a specific, a priori MTDF prediction arising from the finite-range response of the stress field (strain-lever scale $\beta = 22.7$ Mpc). The redshift decay structure is not built into any of the ad-hoc comparison models. Λ CDM does predict environment-correlated SN residuals through peculiar velocities and weak lensing, and Pantheon+ corrects for these using 2M++ reconstructions; the MTDF-specific signature is the excess amplitude, the sharp low- z transition, and its sign.

4.6 Non-parametric confirmation

4.6.1 Permutation test (10,000 shuffles)

Catalogue	p_{perm}
REVOLVER	0.025
VIDE	0.046
VoidFinder	0.062

4.6.2 Block bootstrap (5,000 resamples), 95% CI

Catalogue	95% CI (mag)	Excludes zero?
REVOLVER	[+0.0016, +0.0076]	Yes
VIDE	[+0.0010, +0.0080]	Yes

4.7 Numerical integrity

GPU vs CPU crosscheck: maximum difference = 1.86×10^{-17} (machine epsilon). Results are bit-reproducible.

Phase 3 conclusion. The $\text{SN} \times \text{void}$ environment signal is reproduced by an independent reimplementaion and survives non-parametric robustness checks. Three void catalogues, 16/16 LOSO surveys, permutation tests, and block bootstrap all support a positive γ_{env} with the redshift decay structure predicted by MTDF, and bootstrap 95% confidence intervals exclude zero. This constitutes an independent reimplementaion and non-parametric robustness validation of the MTDF_02 result on the same data. Independent confirmation in the scientific sense requires analysis of new data by external groups, which remains the priority follow-up (Phase 4 forecasts the required sample sizes).

5 Phase 4: Sensitivity forecasts

COMPLETE: 5σ detection at $\sim 3,400$ SNe (LSST/Rubin territory) Phase 4 uses GPU-accelerated Monte Carlo simulations to forecast the sample sizes required for 3σ and 5σ detection of the environment signal under various systematic scenarios. All six consistency checks against Phase 3 pass (relative differences $\sim 10^{-13}$).

5.1 Key structural insight

Rank-1 systematic immunity. A global systematic floor of the form $\sigma_{\text{sys}}^2 \times \mathbf{11}^T$ has *exactly zero* effect on σ_γ . The intercept absorbs global systematics entirely. Only systematics correlated with d_{signed} (the void proximity metric) can degrade the environment coefficient. Phase 3 already bounded these dangerous systematics: survey fixed-effect stability ($\Delta\gamma < 0.0005$), LOSO (16/16 positive), and redshift modulation (signal at $z < 0.04$ only).

The regression is statistics-limited, not systematics-limited.

5.2 Detection threshold table

Finder	Scenario	3σ 50%	3σ 95%	5σ 50%	5σ 95%
REVOLVER	Baseline	1,200	3,000	3,400	6,600
REVOLVER	Realistic	1,200	2,900	3,300	6,600
REVOLVER	Adversarial	>10k	>10k	>10k	>10k
REVOLVER	NGC-only	3,200	7,300	8,300	>10k
VoidFinder	Baseline	1,700	4,500	4,900	9,100
VIDE	Baseline	1,500	4,400	4,500	8,700

5.3 Interpretation

- **Current Pantheon+ (564 SNe):** $\sim 2\sigma$, consistent with Phase 3 ($\Delta\chi^2 = 4.25$ for REVOLVER).
- **Baseline $\approx$ Realistic:** nearly identical thresholds confirm the analysis is statistics-limited, not systematics-limited.
- **Adversarial:** a hypothetical systematic perfectly correlated with d_{signed} would prevent detection at any sample size, but Phase 3 already rules this out empirically.
- **LSST/Rubin (10,000+ low-z SNe):** well into 5σ detection territory for all three void catalogues.

Merged-sample hardening: A merged Pantheon+ / ZTF DR2 / Foundation DR1 sample (4,188 SNe, 3,082 at $z < 0.157$) confirms that the constant- γ global model is null ($\Delta\chi^2 < 1.4$), while a piecewise transition at $z \approx 0.04$ yields $\Delta\chi^2 = 36\text{--}39$ (diagonal covariance; $p < 0.001$) across all three void catalogues. SALT2 population controls shift γ by $< 0.3\sigma$. The signal is localised to the finite-range regime, not a global tilt. See MTFD_02 §2.3 for full merged-sample hardening results.

Phase 4 conclusion. Definitive 5σ detection of the MTFD environment signal requires approximately 3,400 low-redshift supernovae cross-matched with void catalogues (REVOLVER baseline, 50% power). This is within the expected yield of LSST/Rubin. The analysis is statistics-limited: realistic systematics do not materially degrade detection prospects.

6 5b. Phase 3b: Asymmetry and detectability diagnostics

COMPLETE: SGC sensitivity-limited; asymmetry not significant; geometry excludedTo resolve the observed North–South Galactic Cap (NGC/SGC) asymmetry in the Phase 3 environment signal, we implemented an expanded diagnostic battery designed to distinguish physical inconsistency from detectability limits imposed by current survey coverage. All five tests ran in production mode (20s total). All diagnostics reuse the exact Phase 3 environment definition (d_{signed} array) and regression structure. No void assignments, distances, or environment metrics are recomputed.

6.1 5b.1 Test A: IPW matched redshift

After inverse probability weighting to equalise footprint demographics (redshift, host mass proxy, survey origin), no catalogue produces a significant $\Delta\gamma$. All p-values range from 0.36 to 0.99. REVOLVER shows the largest residual at +0.0063, still far from significance. Demographic imbalance between footprints does not explain the Phase 3 signal or its NGC concentration.

6.2 5b.2 Test B: Void–SN joint density mapping

The SGC is severely void-sparse relative to the NGC across all three catalogues:

Catalogue	NGC voids	SGC voids	Ratio
VoidFinder	264	35	7.5×
REVOLVER	511	82	6.2×
VIDE	383	58	6.6×

SN-per-void ratios are 3–8× higher in the SGC, indicating that each SGC void is sampled by far fewer supernovae. This establishes a clear detectability deficit in the SGC independent of any physical signal.

6.3 5b.3 Test C: Wald test and parametric bootstrap

The Wald test for $\Delta\gamma = \gamma_{\text{NGC}} - \gamma_{\text{SGC}}$ returns p-values of 0.49–0.96 across catalogues, indicating no significant asymmetry. The parametric bootstrap (likelihood ratio test under $H_0: \gamma_{\text{NGC}} = \gamma_{\text{SGC}}$) returns $p = 0.17$ –0.24. Neither test rejects the null hypothesis that both footprints share the same underlying γ_{env} .

6.4 5b.4 Test D: Geometry and mode-coupling check

Isotropic mock catalogues preserving the angular mask and redshift selection function produce $\Delta\gamma$ distributions fully consistent with zero. Empirical p-values for the observed $\Delta\gamma$ range from 0.28 to 0.90. Survey geometry alone cannot generate the observed hemispherical difference.

6.5 5b.5 Test E: Null signal injection (SGC recovery)

An NGC-strength γ_{env} signal was injected into SGC supernova distance moduli using the exact Phase 3 d_{signed} values:

Catalogue	Recovery bias	Detection rate at $1\times$ (2σ)	Detection rate at $1.5\times$
REVOLVER	<3%	44%	–
VIDE	–	16%	–
VoidFinder	–	0%	–

VoidFinder’s NGC γ is near zero, so injection at $1\times$ is negligible. REVOLVER is the only catalogue with meaningful SGC recovery power.

REVOLVER recovers the injected signal with less than 3% bias, but achieves only 44% detection rate at 2σ for a $1\times$ NGC-strength injection. This mechanically demonstrates that the SGC lacks the statistical power to detect the signal even when it is present.

6.6 5b.6 Interpretation

Phase 3b conclusion. Three findings, each supported by independent diagnostics:

1. **Not significant.** The NGC/SGC contrast is not statistically significant (Wald $p = 0.49$ –0.96; bootstrap $p = 0.17$ –0.24). After IPW demographic matching, all $\Delta\gamma$ are insignificant ($p = 0.36$ –0.99).
2. **Not geometric.** Isotropic mocks with real mask and selection produce null $\Delta\gamma$ distributions consistent with zero ($p = 0.28$ –0.90). Survey geometry cannot generate the observed pattern.
3. **Sensitivity-limited.** The SGC has 6–8× fewer voids than the NGC. Null injection tests confirm that an NGC-strength signal is recoverable in SGC-like data only at 44% power (REVOLVER, 2σ). The SGC null result reflects insufficient void–SN density, not physical inconsistency with MTFD.

7 Phase 5: Full Planck plik + class_mtdf (superseded by the v2 chains)

v1 RESULT RETIRED (gauge artifact). v2 SEALED: standard parameters unshifted, $k_f = 0.685 \pm 0.500$, $\sigma_8 = 0.8265 \pm 0.0062$ (enhanced growth). $\Delta\chi^2$ vs matched Λ CDM control = -1.29 (bounded best fits). This is a maximum-likelihood comparison, not a Bayes factor or Bayesian evidence calculation.

Phase 5 was the Priority 1 hard falsifier: running the full Planck 2018 plik TTTEEE + lowl TT + lowl EE + lensing likelihood using the `class_mtdf` modified Boltzmann solver. The v1 execution of this phase (minimisation plus converged MCMC) is retained in the archived V74 release and in `output/phase5/` as a historical record. Its numerical results are **void**, for the reason documented below. They are replaced here by the sealed v2 chain results.

Why the v1 numbers are void: the gauge artifact. The v1 `class_mtdf` implementation applied the growth modification μ to the 00 (density) source term in the synchronous gauge. That insertion breaks the near-cancellation between the $k^2\eta$ and $1.5a^2\delta\rho$ terms in the synchronous-gauge Poisson bookkeeping and flips the apparent sign of the growth response. The chains returned $\sigma_8 = 0.790 \pm 0.006$ and $S_8 = 0.802 \pm 0.013$, an apparent clustering suppression read at the time as easing the S_8 tension. The corrected implementation applies the covariant, k -independent $\mu(a)$ in the newtonian gauge (CDM Euler equation), where the modification is gauge-consistent. Under the corrected implementation the growth response is an *enhancement*, as the analytic derivation requires ($\mu > 1$ at all late redshifts). The suppression result, the S_8 -relief reading, the v1 posterior tables, and the v1 robustness numbers built on those chains ($k_f = 0.495/0.50 \pm 0.36$, $\Delta\chi^2 = +0.63$, $\Delta\text{AIC} = +2.63$, $\Delta\text{BIC} = +8.05$) are all downstream of the artifact. They carry no evidential weight.

Permanent methods item (gauge-invariance regression test). Any modified-gravity source term inserted into a Boltzmann solver must be validated by a gauge-invariance regression. Implement the modification independently in synchronous and newtonian gauges, and require agreement of the gauge-invariant growth observables ($f\sigma_8$, $\sigma_8(z)$, C_ℓ) before any chain is run. This test is now a permanent part of the MTDf validation methodology. It is exactly the test that would have caught the v1 artifact before publication.

7.1 Sealed v2 results (corrected implementation)

Apples-to-apples guarantee (unchanged). Both Λ CDM and MTDf configurations use identical likelihood set (plik TTTEEE + lowl TT + lowl EE + lensing), data bins, masks, nuisance list, nuisance priors, cosmological priors, and stop rules. The only differences are the added free parameter k_f and the corrected `class_mtdf` solver in place of standard CLASS.

Run: four chains, ~66 h, frozen flight configuration (hash-pinned binary and YAML). The chains stopped at the frozen hard cap of 200,000 accepted steps per chain, with $R-1 = 0.0085$ (target 0.01, satisfied). $R-1_{\text{cl}} = 0.060$ (strict 0.02 not reached: 95% bounds carry extra sampling noise; means and peaks solid). All interruptions auto-resumed from checkpoint with no sample loss.

Parameter	Λ CDM baseline	MTDF v2	Δ
σ_8	0.8101	0.82651 ± 0.00623	+0.0164 (+1.91 σ , posterior units)
H_0 (km/s/Mpc)	67.382	67.444 ± 0.546	+0.06
Ω_m	—	0.31533 ± 0.00771	negligible
n_s	—	0.96426	negligible
$\ln(10^{10} A_s)$	—	3.03962	negligible
k_f	—	0.685 ± 0.500	0 and 1 both compatible

Sealed v2 reporting set (k_f free), adjudicated by the sealed analysis script under a pre-registered three-posture rule. The adjudication was executed only on the completed chain. The $k_f = 1$ importance-reweighted rider gives the same posture: σ_8 shift $+0.0173$ ($+2.07\sigma$), $f_{\text{kick}} = 0.0033$, $\mu(z_{\text{rec}}) = 1.00002$ (CMB-inert). Native growth-sector units: $S_8 = 0.8473 \pm 0.0138$ (free), 0.8501 ± 0.0132 ($k_f = 1$).

7.2 Interpretation

The corrected covariant $\mu(a)$ enhances late growth by about $+2$ percent, and the full-likelihood chain confirms this end to end. σ_8 lands about 2σ (posterior units) above the Λ CDM baseline. The corrected theory therefore mildly *worsens* the comparison with published weak-lensing clustering amplitudes (sealed posture: S8-WORSENER, both riders agreeing). Standard-parameter shifts are tiny and the CMB-side H_0 is unchanged. What survives unchanged from the v1 chains is the structural finding that Planck alone cannot pin the EFE amplitude. The k_f posterior is broad, with both $k_f = 0$ and the theory value $k_f = 1$ compatible. Planck establishes consistency, not support. Discrimination lives in low-redshift probes.

7.3 Model comparison against the matched Λ CDM control

The matched Λ CDM control chain uses the identical binary, likelihood set, priors, chain profile, and stop rules, with exactly three configuration deltas removing the MTDF theory keys. It reproduces the reference Planck- Λ CDM posterior at full-posterior level, validating the pipeline end to end. Control values: $\sigma_8 = 0.81036 \pm 0.00598$, per-sample $S_8 = 0.83004 \pm 0.01278$, $H_0 = 67.389 \pm 0.544$, with shared-parameter shifts at the 10^{-4} level. At the bounded best fits, MTDF $-\log L = 1385.78$ (at $k_f = 0.476$) against Λ CDM $-\log L = 1386.42$, giving $\Delta\chi^2$ (MTDF $-\Lambda$ CDM) $= -1.29$. The minimisation was seeded from each chain’s best sampled point, and both minimisations converged. Dataset-resolved, the two models agree to about one point on every component (lowl TT 22.29 vs 23.77, lowl EE 396.47 vs 396.27, plik TTTEEE 2343.80 vs 2344.00, lensing 8.99 vs 8.80). The small MTDF gain sits in lowl TT, the late-ISW region. Per-sample growth-sector amplitudes: MTDF $S_8 = 0.8473 \pm 0.0138$ (free) and 0.8501 ± 0.0132 ($k_f = 1$) against the control’s 0.83004 ± 0.01278 .

This is a maximum-likelihood comparison, not a Bayes factor or Bayesian evidence calculation. With one additional free parameter, a one-to-two-point gain is the no-information expectation ($\Delta\text{AIC} = +0.71$, favouring neither model). A BIC-style penalty depends on an adopted effective Planck data count and is not quoted. The full Planck 2018 likelihood does not distinguish the corrected MTDF from Λ CDM. Chain minima as reported by the sampler are minus-log-posterior values and are not compared across models. All numbers above are likelihood-only, from the bounded minima and per-dataset components.

7.4 Scope statement

The v2 run is a full Planck 2018 likelihood test of the corrected MTDF effective cosmological implementation. That implementation is the corrected covariant $\mu(a)$ /EFE sector, with the high-Q carrier represented by its CDM-like limit (standard pressureless CDM as proxy). It is not a Boltzmann implementation of all nine degrees of freedom of the frozen action. The full-action CMB perturbation implementation (unified scalar + vector perturbations) is a named future calculation. Phase-5-level success discharges the CMB confrontation at the effective level only.

8 6b. Phase 6: Cross-probe discriminator tests

COMPLETE: Two discriminators (A, E), three upper limits (B, B2, D), one internal consistency (C), two forecasts (C3, C4), one resolved null (C5), one retired-as-run comparison (C2, v1-posterior-based) - 10 tests total Phase 6 extends validation beyond CMB compatibility toward tests that can *distinguish* MTDF from Λ CDM using independent probes. Ten tests are packaged under `output/phase6/`

, targeting weak lensing, CMB lensing, derived parameter consistency, cross-probe S_8 coherence, growth–environment coupling, BAO residuals, cluster mass ratios, and the $SN \times$ void redshift transition. All code was written from scratch for Phase 6.

Tests A and E positively support MTDF-style predictions (redshift transition, void-specific CMB lensing signal). Test C2, as run against the v1 posterior, is retired (gauge artifact); under the sealed v2 chains the S_8 comparison mildly worsens, and its quantitative revisiting is deferred to the preregistered FWD-S8 programme. Tests B, B2, and D are systematics-clean nulls with explicit upper limits. Test C confirms internal consistency. Tests C3 and C4 quantify sensitivity boundaries. Test C5 resolves the cluster mass ratio ($\eta = 1$). Together they demonstrate: *the pipelines contain no favourable selection and are sound, no hidden contradictions exist, and the boundaries of current discriminating power are mapped.*

8.1 Summary table

Test	Observable	Primary statistic	Result	Implication
A	$SN \times$ void γ_{env} vs z_{cut}	GLS z-score peak over z_{cut} scan	Peak at $z_{\text{cut}}=0.030$: 3.62σ (global $p < 0.005$)	Supports MTDF transition at $z < 0.04$
B	KiDS-1000 γ_t around void centres	$\Delta\gamma_t$ over $R=5\text{--}20$ Mpc/h	$-3.65e-5 \pm 4.68e-5$ (0.8σ); 95%: $ \Delta\gamma_t < 1.2e-4$	Null - bounded, not yet discriminating
B2	KiDS-1000 trough vs ridge γ_t	$\Delta\gamma_{t,\text{split}}$ (ridge–trough)	$+2.56e-5 \pm 5.46e-5$ (0.5σ); 95%: $ \Delta\gamma_{t,\text{split}} < 1.3e-4$	Null - bounded, not yet discriminating
C	Derived parameter consistency	Max shift vs ΛCDM	All within 2.4σ	Internally consistent (v1-posterior-based; the S_8 element is superseded)
D	Planck PR4 $\kappa \times$ DESI BGS voids	Per-void compensated κ_{comp}	$-7.47e-4 \pm 1.03e-3$ (0.7σ , 557 voids)	Null at low z - expected; pipeline validated
E	Planck PR4 $\kappa \times$ BOSS DR12 voids	Conservative κ_{comp} + detection-optimised CTH/MF	Comp: 0.2σ null; CTH: 1.5σ ($p_{\text{rand}}=0.03$); MF: 1.9σ ($p_{\text{rand}}=0.02$)	Pipeline benchmark validated; void-specific signal recovered
C2	S_8 cross-probe comparison (retired as run)	Phase 5 S_8 vs published WL surveys	Combined WL tension: $3.47\sigma \rightarrow 1.89\sigma$ (46% reduction)	Retired as run; under the sealed v2 chains the direction reverses
C3	BAO residual structure	ΔW RMS (wiggle difference MTDF– ΛCDM)	ΔW RMS = 0.27% , max = 0.51% ; sound horizon shift $+0.11\%$	Not detectable by any foreseeable survey (best SNR = 0.92)
C4	$f\sigma_8 \times$ environment	RSD + void $f\sigma_8$ vs $\mu(a)$ prediction	$\Delta\chi^2 = +0.49$ (9 DOF); max discrimination 0.22σ	Sensitivity-limited; needs $10\times$ improvement (DESI/Euclid)
C5	Cluster $M_{\text{dyn}}/M_{\text{lens}}$	Analytical $\mu(a)$ prediction + numerical $\eta(k,z)$ verification	$\eta = 1.000000$ at all cluster scales; Case B confirmed: $M_{\text{dyn}}/M_{\text{lens}} = 1$	Resolved: cluster mass ratio is not a discriminator for MTDF

8.2 6b.1 Test A: Redshift transition scan

- Scans $z_{\text{cut}} \in [0.025, 0.100]$ using GLS and Spearman metrics on the Phase 3 $SN \times$ void environment signal.
- Peak at $z_{\text{cut}} = 0.030$ with GLS z-score 3.62σ ; global scan-corrected permutation $p < 0.005$ (limited by 200 permutations; 0 of 200 exceedances). The scan peak ($z_{\text{cut}} = 0.030$) and the headline transition ($z \approx 0.040 \pm 0.005$) are statistically consistent given the scan’s redshift resolution; they refer to the same low-redshift transition, not two distinct features.
- Spearman rank test agrees independently (3.1σ).

- Interpretation: the environment signal is localised to $z < 0.04$, consistent with the Phase 3 redshift modulation and the MTDF finite-range prediction.
- Artefacts: output/phase6/testA_redshift_transition/

8.3 6b.2 Test B: Weak lensing $\times$ environment (void centres)

- KiDS-1000 shear stacked around DESI REVOLVER void centres and matched random positions; γ_x (cross-shear) gate required to pass before reporting γ_t .
- $\Delta\gamma_t = -3.65 \times 10^{-5} \pm 4.68 \times 10^{-5}$ over $R = 5\text{--}20$ Mpc/h, consistent with zero.
- γ_x gate: pass ($p = 0.234$).
- Explicit two-sided 95% upper limit: $|\Delta\gamma_t| < 1.20 \times 10^{-4}$.
- Without a forward MTDF lensing prediction for this selection, the result is best framed as “not yet discriminating, but now bounded and reproducible.”
- Artefacts: output/phase6/testB_wl_environment/

8.4 6b.3 Test B2: Trough and ridge lensing (projected density split)

- Projected density field built with mask-aware normalisation and full KiDS-1000 footprint (North + South). Both trough and ridge γ_x gates pass.
- $\Delta\gamma_{t,\text{split}}$ (ridge–trough) = $+2.56 \times 10^{-5} \pm 5.46 \times 10^{-5}$ (0.5σ), consistent with zero.
- Explicit two-sided 95% upper limit: $|\Delta\gamma_{t,\text{split}}| < 1.28 \times 10^{-4}$.
- Projected P20/P80 density contrasts ($\delta \sim \pm 0.15$) are too mild for detectable differential lensing at KiDS sensitivity. Spectroscopic 3D density or LSST/DES area would be needed for significance.
- Artefacts: output/phase6/testB2_trough_ridge/

8.5 6b.4 Test C: Derived parameter consistency

- Cross-checks derived parameters from the Phase 5 MCMC posterior against external weak lensing constraints.
- No derived parameter deviates beyond 2.4σ from Λ CDM.
- MTDF’s Planck posterior is internally consistent; no hidden contradictions introduced by `class_mtdf`.
- Artefacts: output/phase6/testC_derived_consistency/

8.6 6b.5 Test D: CMB lensing $\times$ DESI BGS voids (low redshift)

- Planck PR4 MV convergence (κ) stacked on DESIVAST BGS voids ($z < 0.24$) with mask-weighted stacking. Primary statistic is a per-void compensated filter: $\kappa(R/R_v < 0.5) - \kappa(4 < R/R_v < 5)$, designed to remove large-scale mode contamination.
- $\kappa_{\text{comp}} = -7.47 \times 10^{-4} \pm 1.03 \times 10^{-3}$ (0.7σ , 557 voids), consistent with zero.
- RA-scramble null: $p = 0.425$ (compensated). Random-sky null: $p = 0.080$.
- Robustness: stable across outer annulus choices (all $< 1.1\sigma$), low- l removal ($l \geq 20$, $l \geq 30$), and void finder.
- Null at $z < 0.24$ is expected: the CMB lensing kernel peaks at $z \sim 1\text{--}2$, and the BGS void count (557 usable) limits statistical power.

- Diagnosed failure modes from v1: hard mask quality cut rejected 73% of voids; low-l CMB modes created a positive κ pedestal. Both fixed in v2 via per-void compensated filter and mask-weighted stacking.
- Artefacts: output/phase6/testD_cmb_lensing/

8.7 6b.6 Test E: CMB lensing $\times$ BOSS DR12 voids (benchmark)

- Benchmark run on the Mao+2017 ZOBOV catalogue (1,228 BOSS DR12 CMASS+LOWZ voids, $z = 0.2\text{--}0.7$) to validate the CMB lensing stacking pipeline against Cai+2017’s published 3.2σ detection on the same catalogue.
- Five statistics computed side by side:
Per-void compensated (systematics-clean primary): 0.2σ , null by design. **AP disc** (contamination monitor): 4.7σ , but RA-scramble $p = 0.945$ and collapses to 0.3σ under $l \geq 20$ cut - entirely pedestal. **CTH** (Cai-class compensated top-hat): 1.5σ ($p_{\text{rand}} = 0.03$), increases to 2.0σ under $l \geq 20$ cut. **Matched filter** (template + jackknife covariance): 1.9σ ($p_{\text{rand}} = 0.02$), increases to 2.2σ under $l \geq 20$ cut. **Close comp** (alternative aperture): 0.5σ , consistent with zero.
- Detection-optimised statistics (CTH and MF) show the expected hierarchy and respond correctly to pedestal removal and footprint-preserving null tests. The gap from $\sim 1.5\text{--}2.2\sigma$ to Cai+2017’s 3.2σ reflects template shape (step vs theory-derived) and different Planck products (PR4 vs PR2015).
- Artefacts: output/phase6/testE_boss_benchmark/

8.8 6b.7 Test C2: S_8 cross-probe comparison (RETIRED as run; direction reversed under v2)

- **As-run status: RETIRED.** Test C2 compared the v1 Phase-5 posterior ($S_8 = 0.802 \pm 0.013$) against published weak-lensing constraints and reported a tension reduction. The v1 posterior was a gauge artifact (Section 6), so the tension-reduction result is void.
- **Under the sealed v2 chains the direction reverses:** the corrected implementation gives an enhanced-growth posture ($\sigma_8 = 0.8265 \pm 0.0062$, about $+1.9\sigma$ posterior units above the Λ CDM baseline; native $S_8 = 0.8473 \pm 0.0138$), which mildly *worsens* the comparison with published weak-lensing compressions. No relief claim survives.
- The as-run caveat stands and is now load-bearing: published WL S_8 values assume Λ CDM growth and lensing kernels, so the only pre-registered path for quantitatively revisiting this comparison is the forward-modelling programme (FWD-S8), which forward-models each survey’s compression within MTDf rather than comparing native posteriors to Λ CDM-compressed numbers.
- Artefacts (v1, historical): output/phase6/testC2_s8_coherence/

8.9 6b.8 Test C3: BAO residual structure

- Extracts BAO wiggles $W(k) = P(k)/P_{\text{nw}}(k) - 1$ from `class_mtdf` power spectra using an Eisenstein & Hu (1998) no-wiggle reference, and computes the residual $\Delta W = W_{\text{MTDF}} - W_{\Lambda\text{CDM}}$.
- Sound horizon shift: $+0.106\%$ ($147.148 \rightarrow 147.303$ Mpc). Wiggle difference: ΔW RMS = 0.27% , max = 0.51% at $k = 0.12$ h/Mpc. No detectable peak phase shift at grid resolution.
- Best projected SNR: 0.92 (DESI + Euclid combined). The 0.1% sound horizon shift is absorbed into standard cosmological parameter degeneracies.

- Conclusion: BAO residual structure is not a viable MTDF- Λ CDM discriminator with any foreseeable survey.
- Artefacts: output/phase6/testC3_bao_residuals/

8.10 6b.9 Test C4: $f\sigma_8 \times$ environment

- Compares MTDF and Λ CDM $f\sigma_8(z)$ predictions against a compilation of 8 RSD surveys plus void-specific growth measurements from Hamaus+2020.
- χ^2 : Λ CDM = 7.61, MTDF = 8.10 ($\Delta\chi^2 = +0.49$, 9 DOF). Maximum per-point discrimination: 0.22σ .
- The homogeneous $\mu(a)$ modification produces a $\sim 1.5\%$ signal, which is $5\text{--}10\times$ smaller than current RSD measurement errors (0.03–0.10 per z -bin).
- Future requirement: 2σ discrimination needs $\sigma(f\sigma_8) \sim 0.003\text{--}0.004$ per z -bin, achievable with DESI/Euclid-era RSD or environment-resolved estimators.
- Artefacts: output/phase6/testC4_fsiga8_environment/

8.11 6b.10 Test C5/C5b: Cluster $M_{\text{dyn}}/M_{\text{lens}}$ - Resolved

- C5 computed the predicted ratio $M_{\text{dyn}}/M_{\text{lens}}$ under two bracketing coupling cases:
Case A (μ -only dynamics, $\Sigma = 1$): $M_{\text{dyn}}/M_{\text{lens}} = 1.053$ at $z = 0$ (+5.3% offset). **Case B** (equal coupling, $\Sigma = \mu$): $M_{\text{dyn}}/M_{\text{lens}} = 1.000$ (no offset).
- **C5b resolved the key unknown** by extracting $\eta(k,z) = \Phi(k,z)/\Psi(k,z)$ directly from `class_mtdf` Newtonian-gauge transfer functions across $k = 0.005\text{--}2$ h/Mpc and $z = 0\text{--}10$.
- Result: $\eta = 1.000000$ at all cluster-relevant scales ($z \leq 0.8$). Max $|\Delta\eta|$ between MTDF and Λ CDM: 0.000000. Tiny deviations ($|\eta-1| \sim 10^{-5}$) appear only at $z \geq 5$ from standard radiation anisotropic stress, present equally in both models.
- Physical reason: `class_mtdf` modifies only the Poisson equation source ($\delta\rho \rightarrow \mu\delta\rho$) and the CDM Euler equation. The $\Phi\text{--}\Psi$ anisotropy equation is unchanged from GR. Both potentials are enhanced equally $\rightarrow \Sigma = \mu \rightarrow$ **Case B confirmed**.
- Implication: $M_{\text{dyn}}/M_{\text{lens}} = 1.000$ under MTDF. The cluster mass ratio channel is **not a discriminator**. Clusters can still test MTDF through growth-dependent observables (e.g. cluster counts), because $\mu > 1$ modifies the growth rate.
- Artefacts: output/phase6/testC5_cluster_mass/, output/phase6/testC5b_gravitational_slip/

Phase 6 conclusion (10 tests, all complete). The ten Phase 6 tests fall into five categories. *Discriminators* (6A, 6E): the redshift transition is identified at 3.6σ ; the BOSS CMB lensing benchmark recovers void-specific structure at $1.5\text{--}1.9\sigma$. *Coherence* (6C, 6C2): derived parameters are self-consistent (no shift $> 2.4\sigma$); the as-run 6C2 S_8 comparison is retired with the v1 chains (direction reversed under v2). *Upper limits* (6B, 6B2, 6D): systematics-clean nulls with explicit 95% bounds on void lensing and CMB lensing signals. *Forecasts* (6C3, 6C4): BAO wiggles and $f\sigma_8$ quantify what future surveys need to reach. *Resolved* (6C5): cluster $M_{\text{dyn}}/M_{\text{lens}}$ is null ($\eta = 1$). No surviving test contradicts MTDF at the consistency level. The strongest positive outcomes are the discriminators (6A redshift transition, 6E void-specific CMB lensing structure); the former 6C2 relief claim does not survive the v2 correction and is retired.

9 Synthesis and framework narrative

9.1 Summary of results (17/17 phases complete)

Phase	Type	Status	Key result	Artefacts
Phase 1	Discriminator	PASSED	V18 strict totals reproduced to $\delta = 0.000049$	output/phase1/
Phase 2	Discriminator	PASSED	CMB plik-lite compatible ($\Delta\chi^2 < 1$); $k_f = 1$ within 95% in all MCMC runs	output/phase2/
Phase 3	Discriminator	PASSED	$SN \times$ void signal reproduced exactly; bootstrap excludes zero	output/phase3/
Phase 3b	Upper limit	COMPLETE	NGC/SGC asymmetry: not significant, not geometric, sensitivity-limited (5 diagnostics)	output/phase3b/
Phase 4	Forecast	COMPLETE	5σ at 3,400 SNe (LSST/Rubin); statistics-limited	output/phase4/
Phase 5 (v1)	Discriminator	RETIRED	v1 minimisation and MCMC results void (gauge artifact, Section 6); archived as historical record.	output/phase5/
Phase 5 (v2, sealed)	Discriminator	PASSED (consistency)	Corrected implementation, $R-1 = 0.0085$. $\sigma_8 = 0.8265 \pm 0.0062$ (enhanced growth, $+1.91\sigma$ posterior units above Λ CDM baseline); $k_f = 0.685 \pm 0.500$, both 0 and 1 compatible. $\Delta\chi^2$ vs matched control complete (see Section 6.3).	
Phase 6A	Discriminator	PASSED	Redshift transition scan: signal peaks at $z < 0.03$ (3.6σ GLS, global $p < 0.005$). Spearman confirms (3.1σ).	
Phase 6B	Upper limit	COMPLETE	Weak lensing $\times$ void environment (KiDS-1000): $\Delta\gamma_t = -3.65e-5 \pm 4.68e-5$ (0.8σ). γ_x gate pass. 95% UL: $ \Delta\gamma_t < 1.2e-4$.	output/phase6/
Phase 6B2	Upper limit	COMPLETE	Trough/ridge lensing (KiDS-1000): $\Delta\gamma_{t,split} = +2.56e-5 \pm 5.46e-5$ (0.5σ). Both γ_x gates pass. 95% UL: $ \Delta\gamma_{t,split} < 1.3e-4$.	
Phase 6C	Coherence	PASSED	Derived parameter consistency: no shifts $> 2.4\sigma$ (v1-posterior-based; the S_8 element is superseded, see 6b.7).	
Phase 6C2	Coherence	RETIRED (as run)	Compared the v1 Phase-5 posterior against WL surveys; the v1 posterior is void (gauge artifact), so the reported tension reduction is retired. Direction reversed under sealed v2; quantitative revisit deferred to FWD-S8.	

Phase 6C3	Forecast	COMPLETE	BAO residual structure: ΔW RMS = 0.27%, sound horizon shift +0.11%. Not detectable by any current or projected survey (best SNR = 0.92).
Phase 6C4	Forecast	COMPLETE	$f\sigma_8 \times$ environment: RSD + void growth vs $\mu(a)$ prediction. Max 0.22σ ; needs DESI/Euclid-era precision ($10\times$ improvement).
Phase 6C5/C5b	Closed null	RESOLVED	Cluster $M_{\text{dyn}}/M_{\text{lens}}$: $\eta = 1$ confirmed numerically (C5b) $\Rightarrow \Sigma = \mu$, so $M_{\text{dyn}}/M_{\text{lens}} = 1$ exactly. Mass ratio offset is zero in the current MTFD perturbation model.
Phase 6D	Upper limit	COMPLETE	CMB lensing $\times$ DESI BGS voids: $\kappa_{\text{comp}} = -7.47\text{e-}4 \pm 1.03\text{e-}3$ (0.7σ , 557 voids). RA-scramble clean. Null expected at $z < 0.24$.
Phase 6E	Discriminator	COMPLETE	CMB lensing $\times$ BOSS DR12 benchmark: CTH 1.5σ ($p_{\text{rand}}=0.03$), MF 1.9σ ($p_{\text{rand}}=0.02$). Pipeline validated against Cai+2017.

9.2 Claim status taxonomy

Each MTFD claim or test outcome is assigned one of four status labels. This taxonomy is intended to provide referees and readers with an unambiguous assessment of where the framework stands.

Label	Definition
✓ Validated	Independently reproduced with quantitative agreement. Robust to systematics tests.
■ Constrained	Consistent with data but not uniquely favoured. Current data cannot distinguish from Λ CDM.
▲ Hypothesis	Theoretically motivated but not yet tested against data, or tested only indirectly.
× Falsifier	Identified test that could refute the framework. Not yet executed.

9.2.1 Status by claim

Claim / element	Status	Evidence
V18 strict χ^2 totals	✓ Validated	Phase 1: independent reproduction to $\delta = 0.000049$
Pantheon+ SN likelihood	✓ Validated	Phase 1: $\chi^2 = 2012.72$ matched exactly
DESI Y1 BAO fit	✓ Validated	Phase 1: $\chi^2 = 15.39$ matched exactly
Cosmic chronometers $H(z)$	✓ Validated	Phase 1: $\chi^2 = 6.98$ matched exactly
DR16 $f\sigma_8$ growth	✓ Validated	Phase 1: $\chi^2 = 2.00$ matched exactly
CMB compatibility (plik-lite)	■ Constrained	Phase 2: $\Delta\chi^2 < 1$; $k_f = 1$ within 95% CI. Perturbative correction layer, not full Boltzmann.

CMB compatibility (full plik + class_mtdf)	✓ Validated	Sealed v2 chains (corrected implementation): standard parameters essentially unshifted; $k_f = 0.685 \pm 0.500$ with 0 and 1 both compatible; apples-to-apples configuration; $\Delta\chi^2$ vs matched Λ CDM control complete (see Section 6.3). (v1 Phase-5 numbers retired, gauge artifact.)
SN $\times$ void environment signal (γ_{env})	✓ Validated	Phase 3: exact reproduction, 3 void finders, bootstrap excludes zero
Redshift decay structure ($z < 0.04$)	✓ Validated	Phase 3: a priori prediction matched ($p = 0.0035$ REVOLVER)
Survey systematics stability	✓ Validated	Phase 3: FE $\Delta\gamma < 0.0005$; LOSO 16/16 positive
NGC/SGC asymmetry interpretation	✓ Validated	Phase 3b: asymmetry not significant (Wald $p = 0.49$ – 0.96), not geometric (mock $p = 0.28$ – 0.90), SGC sensitivity-limited (6–8 $\times$ fewer voids, 44% recovery power). Five independent diagnostics confirm footprint-dependent detectability.
5σ detection forecast (3,400 SNe)	✓ Validated	Phase 4: Monte Carlo with consistency checks passing at 10^{-13}
Statistics-limited (not systematics-limited)	✓ Validated	Phase 4: baseline $\approx$ realistic; rank-1 systematic immunity demonstrated
Hubble tension consistency (conditional; void depletion)	■ Constrained	Void-depletion mapping yields H_{local} consistent with SH0ES, conditional on local-underdensity depth. Mechanism independent of CMB; not yet tested with dedicated void-SN joint analysis.
Photon-stress coupling ($\kappa = f_{\text{kick}}/3 \approx 0.00102$)	▲ Hypothesis	Derived from fundamental parameters (angular averaging of strain tensor). Below Planck sensitivity. Paper MTDF_04 frames as speculative.
Rotation curve phenomenology	■ Constrained	Included in V18 scalars. Not independently reproduced within this programme.
Redshift transition structure (z_{cut} scan)	✓ Validated	Phase 6A: GLS z-score 3.62σ at $z_{\text{cut}}=0.030$; global scan-corrected $p < 0.005$. Spearman confirms (3.1σ).
Weak lensing $\times$ environment (KiDS)	■ Constrained	Phase 6B/B2: systematics-clean nulls with explicit 95% upper limits. Bounded but not yet discriminating at KiDS sensitivity.
Derived parameter consistency	✓ Validated	Phase 6C: all within 2.4σ (v1-posterior-based; S_8 element superseded, see the row below).
CMB lensing $\times$ voids	■ Constrained	Phase 6D/E: low-z null (expected), BOSS benchmark recovers void-specific structure. Pipeline validated.
S_8 cross-probe comparison	$\times$ Retired (as run)	Phase 6C2 compared the v1 Phase-5 posterior against WL surveys; that posterior is void (gauge artifact). Under the sealed v2 chains the direction reverses (enhanced growth, S8-WORSENERD); quantitative revisit deferred to FWD-S8.
$f\sigma_8 \times$ environment	■ Constrained	Phase 6C4: $\mu(a)$ prediction vs RSD + void growth. Max 0.22σ discrimination; current data sensitivity-limited (10 $\times$ improvement needed).
BAO residual structure	■ Constrained	Phase 6C3: ΔW RMS = 0.27%. Sound horizon shift +0.11% absorbed into parameter degeneracies. Not detectable by any foreseeable survey.
Cluster $M_{\text{dyn}}/M_{\text{lens}}$	■ Constrained	Phase 6C5/C5b: $\eta = 1.000000$ confirmed numerically from class_mtdf transfer functions. Case B applies: $M_{\text{dyn}}/M_{\text{lens}} = 1$. Mass ratio channel closed as discriminator; growth-dependent cluster tests remain open.

Weak lensing P(k) cross-correlation	× Falsifier	Not yet executed. Requires full P(k) from class_mtdf at non-linear scales.
Merging cluster offsets (Bullet Cluster)	× Falsifier	Not yet executed. Requires N-body simulations.

9.3 Framework narrative

MTDF is fully CMB-compatible: Planck TTTEEE cannot distinguish MTDF from Λ CDM ($\Delta\chi^2 < 1$). MTDF deviates from Λ CDM at late times through environment-dependent dynamics. The Hubble tension is consistent with environmental mapping (void stress depletion), conditional on the depth of the local underdensity, rather than with early-universe modification. The $\text{SN} \times \text{void}$ signal is the primary empirical discriminator. The finite range of the stress-field response concentrates observable effects at low redshift, consistent with the CMB being unaffected.

9.4 What survives intact

1. **Phase 1 strict totals:** independently reproduced to four decimal places.
2. **$\text{SN} \times \text{void environment signal}$:** three void finders, 16/16 LOSO, redshift decay matching a priori prediction.
3. **Hubble tension consistency (conditional):** entirely late-time, via void stress depletion:
 Theoretical: $H_{\text{local}}/H_0 = 1.084 \Rightarrow 67.4 \times 1.084 = 73.1$ (SH0ES: 73.04 ± 1.04), conditional on local-underdensity depth (shallow void gives ~ 70 – 71) MCMC-derived (sealed v2 chains): $H_{0,\text{global}} = 67.44 \pm 0.55$ km/s/Mpc; applying the void-mapping factor $S_0 = 1.084$ gives a local-environment value $H_{0,\text{local}} \approx 73.1$ km/s/Mpc, conditional on the depth of the local underdensity. The relation $S_0 = \alpha\sqrt{\Omega_{\text{DE}}}$ is an algebraic identity, not an independent prediction (see MTDF_01 §3.3).
4. **Late-time phenomenology:** $w(z)$, BAO distances, growth rate, rotation curves.
5. **Framework stability:** MTDF recovers Λ CDM at early times, deviates at late times.
6. **Cross-probe integrity (Phase 6, 10 tests):** redshift transition identified (3.6σ); weak lensing and CMB lensing pipelines pass strict systematics controls; derived parameters self-consistent; $f\sigma_8$, BAO residuals, and cluster mass predictions computed - all sensitivity-limited with current data, with quantified thresholds for future surveys.

9.5 Joint implications for late-time cosmological tensions

While MTDF is not tuned to resolve any single cosmological tension, the validated framework exhibits simultaneous shifts in multiple late-time observables. In particular, the Phase 5 best-fit favours a modest reduction in σ_8 relative to Λ CDM best-fit values (0.795 vs 0.809), consistent with the direction preferred by weak-lensing surveys (KiDS, DES).

Independently, the environment-dependent dynamics tested in Phases 3 and 4 naturally modify local expansion observables without compromising global CMB consistency. Together, these results indicate that MTDF addresses multiple late-time phenomenological tensions within a single, unified dynamical framework, without introducing additional free parameters beyond k_f .

9.6 Paper implications

Paper	Impact
MTDF_01 (Master)	Phase 1 strengthens several claims. MTDF_01 updated in the v1.1.6 claims audit (see CHANGELOG).
MTDF_02 (Part I, Environmental)	Signal reproduced by an independent reimplementation and non-parametric robustness checks (Phase 3). Redshift transition near $z \approx 0.04$, identified at 3.6σ (Phase 6A, scan-corrected $p < 0.005$); the merged-sample break is not sharply localised at this value. Framework’s strongest empirical evidence to date.
MTDF_04 (Part III, Photon coupling)	$\kappa \approx f_{\text{kick}}/3$ is a theory-motivated value, consistent with but not measured by FIRAS bounds, structurally related to the early field energy fraction and below Planck sensitivity. Paper frames this as speculative, which is appropriate.
MTDF_05 (Part IV, Cosmological)	Sealed v2 full-Planck results govern (Section 6.1); Phase 2 plik-lite precursor (+0.81) and the CMB+BAO summary run (+1.25, narrower data) are pre-correction bookkeeping. Hubble tension mechanism intact, conditional on local-underdensity depth. $\Delta\chi^2$ vs matched control complete (see Section 6.3).
MTDF_07 (This paper)	Phase 6 adds ten cross-probe tests (6A–6E, 6C2–6C5). Title updated to Phases 1–6. All validation claims current as of February 2026.

10 Open questions and next steps

10.1 NGC/SGC asymmetry diagnostics (Phase 3b, complete)

Phase 3b (Section 5b) implemented the full asymmetry diagnostic battery and is now complete. All five tests confirm that the NGC/SGC contrast is not statistically significant, not caused by survey geometry, and attributable to SGC void-catalogue sparsity. The NGC/SGC claim status in the taxonomy (Section 7.2) has been upgraded from **▲ Hypothesis** to **✓ Validated**.

10.2 Falsifier and discriminator tests

#	Test	Type	Status	Timescale
1	Full Planck plik + class_mtdf	Discriminator	PASSED at consistency level (sealed v2 chains; $\Delta\chi^2$ vs matched control complete, see Section 6.3)	v2 and control chains sealed
2	NGC/SGC diagnostic battery	Upper limit	COMPLETE (sensitivity-limited, not geometric)	Complete
3	Weak lensing $\times$ environment (KiDS)	Upper limit	COMPLETE (Phase 6B/B2: systematics-clean null)	Complete
4	CMB lensing $\times$ voids	Upper limit	COMPLETE (Phase 6D/E: pipeline validated)	Complete
5	S_8 cross-probe comparison	Coherence	RETIRED as run (v1-posterior-based; direction reversed under sealed v2)	Deferred to FWD-S8
6	$f\sigma_8 \times$ environment	Forecast	COMPLETE (Phase 6C4: 0.22σ , sensitivity-limited)	Complete
7	BAO residual structure	Forecast	COMPLETE (Phase 6C3: $\Delta W = 0.27\%$, undetectable)	Complete
8	Cluster $M_{\text{dyn}}/M_{\text{lens}}$	Resolved	RESOLVED (Phase 6C5/C5b: $\eta = 1$, Case B, $M_{\text{dyn}}/M_{\text{lens}} = 1$)	Complete
9	Weak lensing $P(k)$ (non-linear)	Discriminator	Requires full $P(k)$ from class_mtdf	Months
10	Merging cluster offsets	Discriminator	Requires N-body simulations	Institutional

Phase 2 scope note, resolved by Phase 5. Phase 2 used a perturbative correction layer on a Λ CDM emulator, not the full modified Boltzmann equations. The full calculation now on record is the sealed v2 chain (Section 6.1), which confirms Phase 2’s structural finding (CMB-level consistency; k_f unconstrained by Planck) under the corrected implementation; Phase 2’s and the v1 Phase 5’s numerical $\Delta\chi^2$ values are historical.

11 Appendix: Technical notes

11.1 A.1 k_f normalisation conventions

Three different k_f conventions were identified across the MTFD codebase during Phase 2. Understanding these is essential for interpreting historical results.

Convention	"Full MTFD" k_f	f_{kick}	Source
A (old CLASS, pre-correction)	0.10	0.003303	November 2025 cobaya runs
B (corrected CLASS)	1.0	0.003303	Theory prediction
C (Phase 2 code, initial, corrected)	0.001326	0.001326	CosmoPower correction layer (bug)

Convention C initially used $f_{\text{kick}} = \alpha\kappa = 0.001326$ instead of the correct $f_{\text{kick}} = \lambda_{\text{MTDF}}/24 = 0.003303$. This factor-of-2.49 error caused a $3\times$ overcorrection in the angular scale ratio per unit k_f , leading to a spurious H_0 - k_f degeneracy in the first MCMC run (best-fit $H_0 = 55.6$, which is unphysical). After correction, all runs show $H_0 \approx 67$ and no degeneracies.

11.2 A.2 V18 parameter reference

Parameter	Value	Uncertainty	Role
α	1.30	0.26	Stress-field coupling (void dynamics: KBC 2013 + Hoscheit 2018; counter: Kenworthy 2019)
β	22.7 Mpc	0.09×10^{23} m	Strain-lever length scale (void radius distribution: Pan+2012; Hamaus+2014)
τ	13.0 Gyr	0.2	Field age
β_{eos}	0.573	0.012	Equation-of-state parameter
E	9.1×10^{-10} Pa	0.4×10^{-10}	Field energy scale
δ_{bf}	0.150	0.030	Fixed post hoc from cluster total baryon fractions at $M_{500} \sim 10^{14} M_{\odot}$; a first-principles derivation is not claimed (Giordini+2009; Gonzalez+2013; see MTFD_01 §5.1)
z_t	0.74	0.10	Transition redshift
κ	0.00102	0.00003	Photon–stress coupling (theory-motivated: $\kappa \approx f_{\text{kick}}/3$; below FIRAS sensitivity)
H_0	70.0 km/s/Mpc	1.5	Hubble constant (Freedman 2021 TRGB: 69.8 ± 0.6)
f_{kick}	0.003303	–	Derived: $(1 - \beta_{\text{eos}})^2 / ((1 + \alpha) \times 24)$

11.3 A.3 Historical context: November 2025 runs

Previous `cobaya` + `class_mtdf` runs (November 2025) provide useful reference points. A single-point evaluation at the Planck Λ CDM best-fit gave $\chi^2 = 614.23$ (consistent with full `plik` TT+TE+EE). An MCMC with Planck TT-only (16 walkers, 200 steps) found $k_f \approx 0.03$ in the old normalisation (Convention A), with χ^2 indistinguishable from Λ CDM. These early results are consistent with the current Phase 2 conclusion that Planck cannot constrain the EFE amplitude.

12 Data and code availability

The complete MTFD reproducibility package, including the modified CLASS solver (`class_mtdf`), the V19 strict validation workbook, the `run_validate.py` pipeline, the gravity-sector artefact manifest, and the Phase 2 CosmoPower emulator cache, is archived on Zenodo (concept DOI 10.5281/zenodo.19741058; this release, v1.1.7, DOI 10.5281/zenodo.21525855) and mirrored on GitHub at github.com/IMesche/MTDF. The v1.1.7 Git tag corresponds to the archived Zenodo release. All numerical results reported here can be regenerated from the shipped workbook and scripts following the procedure in the top-level `README.md`.

13 References

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